
Applied Financial Analytics Using Data Science for Business

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01 ✨ Introduction ✨



Project Goal

Our goal for the project:

- Determine which **variables** within companies' financial information **influence profitability** the most
- Recognize trends
- Build a **regression model** to make **predictions** for **future profitability**



02



Dataset



Dataset Background

- Dataset from Alpha Vantage
 - Used API to retrieve data
- Dataset Contains 3 parts
 - Income Statement
 - Cash Flow Statement
 - Balance Sheet
- Companies Chosen
 - 6 Total Companies

```
import requests
import pandas as pd
from google.colab import files

# Key and Stock Symbol
ALPHA_KEY = "KHAVEXTP86XHWU5" # API Key
SYMBOL = "MSFT"

# Annual Reports
def get_financials(function_name, symbol, key, filename):
    url = f"https://www.alphavantage.co/query?function={function_name}&symbol={symbol}&apikey={key}"
    data = requests.get(url).json()
    if 'annualReports' in data:
        df = pd.DataFrame(data['annualReports'])
        df.columns = [c.strip() for c in df.columns]
        df.to_csv(filename, index=False)
        print(f"{filename} saved!")
    else:
        print(f"Error fetching {function_name}: {data}")
    return
```



Features in the Dataset

Balance Sheet

- Date
- Currency Type
- Total Assets
- Investments
- Inventory
- Liabilities
- Debt
- Equity

Income Statement

- Date
- Currency Type
- Gross Profit
- Total Revenue
- Total Costs and Expenses
- Taxes
- Depreciation
- Net Income

Cash Flow Statement

- Date
- Currency Type
- Operating Cash Flow
- Capital Expenditures
- Cash Flow From Investment







03 ✨ Data Cleaning ✨



Data Cleaning



How did we Clean the Dataset

1. Formatting

The raw financial CSV files were first standardized to ensure consistency across datasets. Column names were cleaned and unified, and financial figures originally stored as text were converted into numeric format.

- Special characters such as commas, currency symbols, and extraneous formatting were removed to allow accurate numerical operations. These steps created a consistent and machine-readable structure across all financial statements.

2. Quality Control

- To improve data reliability, missing values were carefully examined and handled to avoid introducing bias into the analysis.
- Duplicate fiscal years were identified and removed to prevent double-counting, and the remaining records were sorted chronologically.
- These quality control measures ensured that the time-series data accurately reflected each firm's financial history.

3. Integration

- After cleaning, the balance sheet, income statement, and cash flow statement were merged into a single consolidated dataset. The merge was performed using fiscal year alignment to maintain consistency across financial reports.
- This integration provided a comprehensive view of each firm's financial performance and position within a unified analytical framework.

4. Feature Engineering

- With the cleaned and merged dataset, key financial ratios were engineered to support regression modeling.
- These ratios transformed raw accounting figures into economically meaningful variables, improving comparability across firms and time periods. The engineered features served as inputs for statistical analysis, enabling more robust interpretation of financial relationships and model results.

Data Preprocessing & Feature Engineering

Methods used:

01

Data Type Harmonization

02

Missing Value Handling

03

Duplicate Record Removal

04

Column Name Standardization

05

String Cleaning & Sanitization

06

Chronological Data Ordering



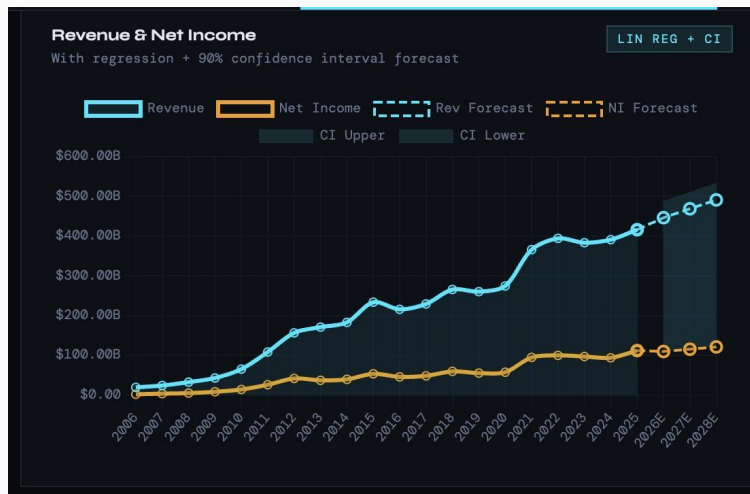
04



EDA



Apple (AAPL)



Years:2006–2012: consistent growth.
 2013–2016: growth slows slightly.
 2020–2022: large increase.
 After 2022: stable upward trend.
 (Strong trend with low volatility)

Gross Margins: Increased from 29%-46%
 Operating Margins: 12% to 33%
 Net Margins: 12% to 27%(efficiency improvement)



EBITDA

With forecast + CI

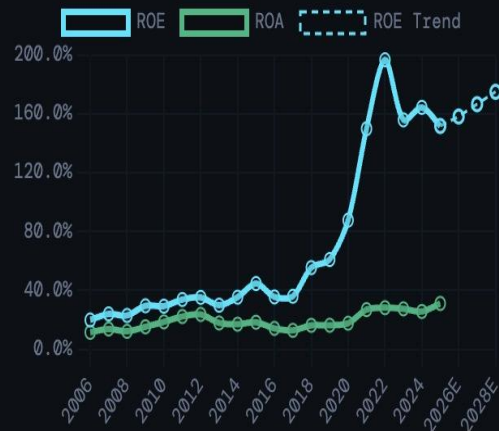
EBITDA



ROE & ROA

Return on Equity & Assets

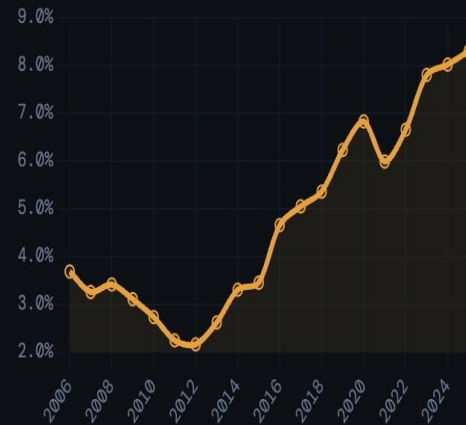
RETURNS



R&D Spend

As % of Revenue

R&D



Increasing
CI forecast widens moderately
Strong positive correlation with
revenue (likely $r > 0.95$)

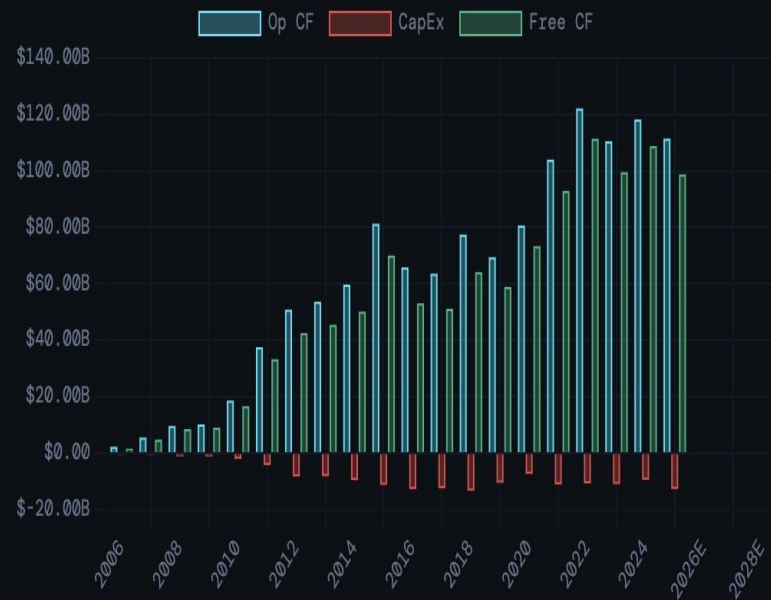
ROE: Massive spike around 2021 (~200%)
Then stabilizes around 150–170%
Highly volatile
ROA: more stable (10%–30%) better
indicator for profit

Decline then strongly increase
(effective in investment)
No negative margin effect

Free Cash Flow

Operating CF · CapEx · FCF with forecast + CI

FCF + CI

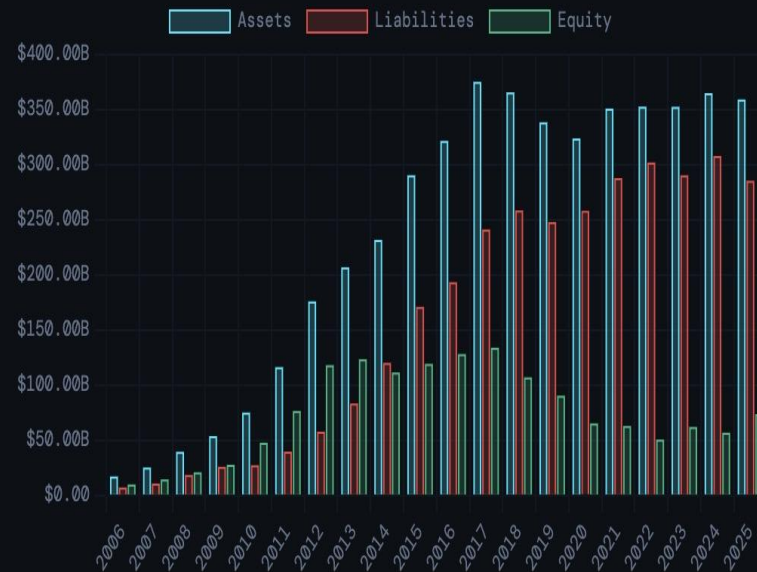


Op CF consistently positive
CapEx moderate,
FCF strong positive since 2012
Funding expansion(high cash conversion ratio)

Capital Structure

Assets · Liabilities · Equity

BALANCE SHEET



Assets increasing
Liabilities increasing
Equity drop after 2018(same as ROE)
Financial engineering component present
Not purely operational growth



Current Ratio

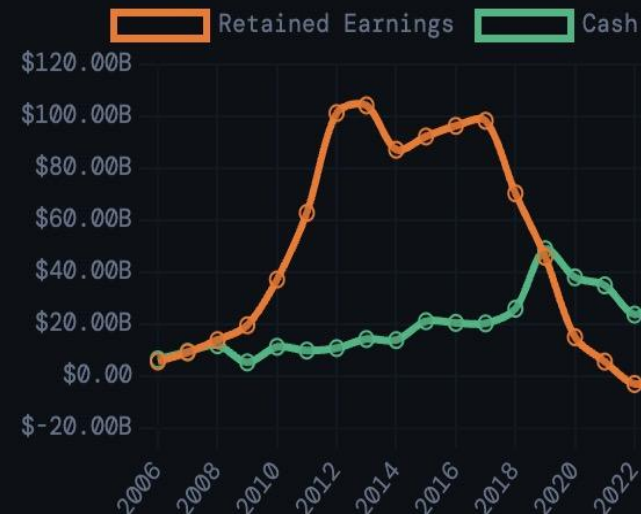
Short-term liquidity



- 2007: 2.2x 2025: 0.9x
- Liquidity tightening(ratio<1)
- Negative time trend
- Structural liquidity shift

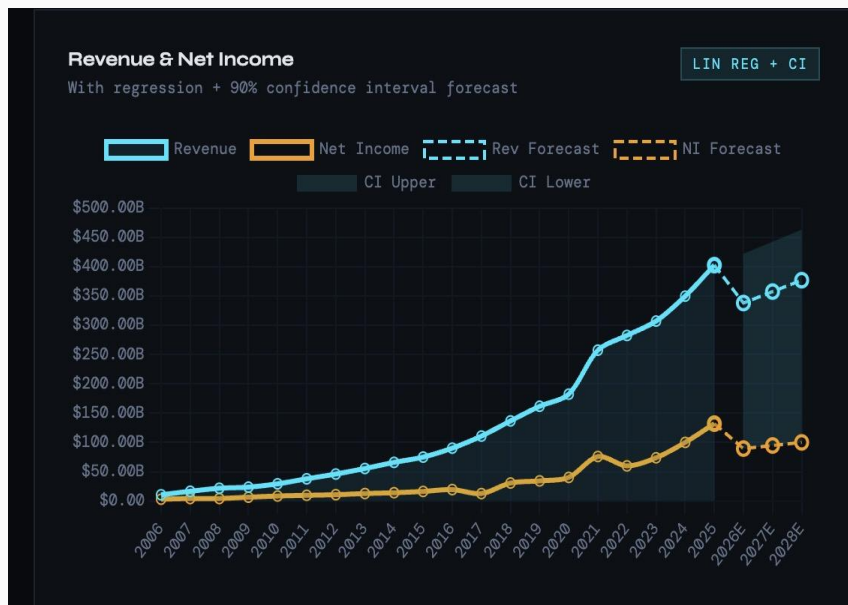
Retained Earnings & Cash

Value accumulation over time



- Retained Earnings: Rise until ~2018
- Then sharp decline into negative
- Earnings distributed aggressively
- Possibly buybacks > net income
- Capital return strategy shift post-2018

Google (GOOG)



Revenue Strong and steady growth from 2006 to 2025.
 Net income gradual growth except drop in 2022. Small drop in forecasting
 Remains profitable



Gross Margin between 55%–65%.
 Operating Margin Around 30% early years.
 Net Margin is volatile
 Consistently profitable with temporary disruptions



EBITDA

With forecast + CI

EBITDA



Strong upward trend.
Big increase after 2020.
cash earnings are strong
Clear Growth trend

ROE & ROA

Return on Equity & Assets

RETURNS



ROE between 15%-30%
ROA around 15%-22%(more stable)
Strong returns on capital.
No extreme volatility.
Performance improved after 2020.

R&D Spend

As % of Revenue

R&D

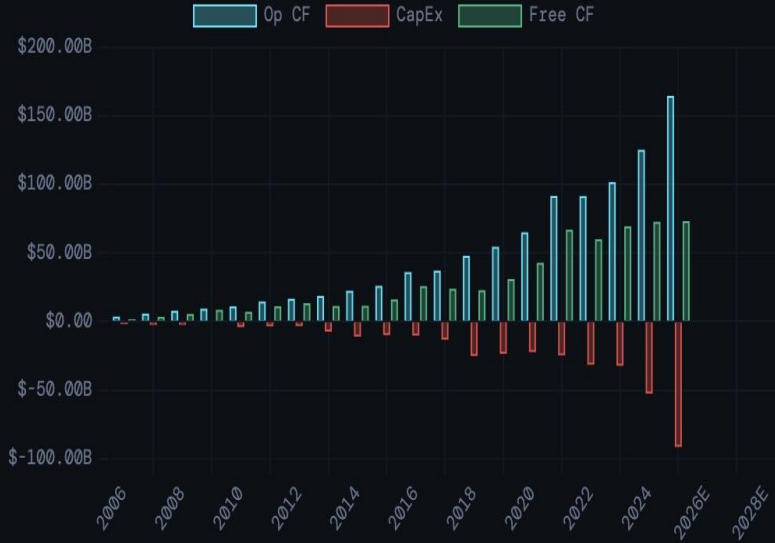


R&D is between 11%–16%
Multiple peaks stable at 15%
invests heavily in innovation.

Free Cash Flow

Operating CF · CapEx · FCF with forecast + CI

FCF + CI



CF: Very strong growth over time.

Large increase after 2021

CapEx: increased sharply in recent years

FCF: remains positive

Company generates strong cash flow.

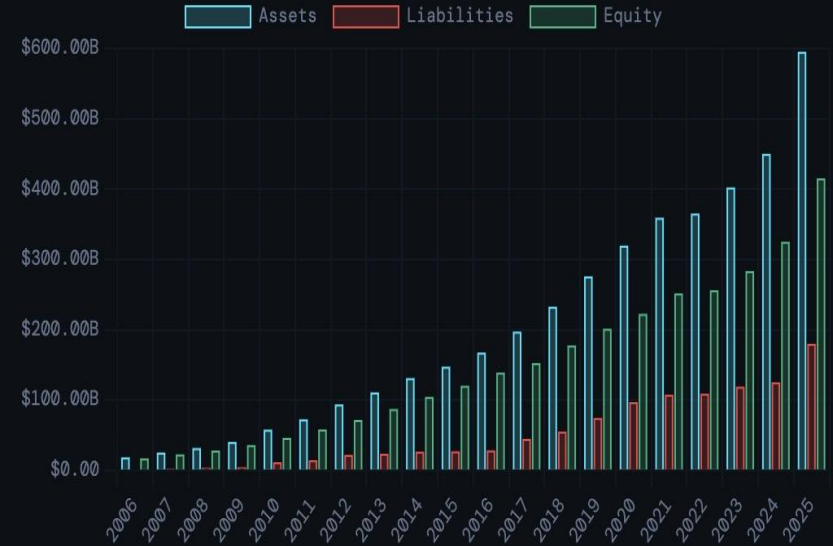
Recently increasing investment



Capital Structure

Assets · Liabilities · Equity

BALANCE SHEET



Assets increasing rapidly.

Liabilities increasing but much lower than assets.

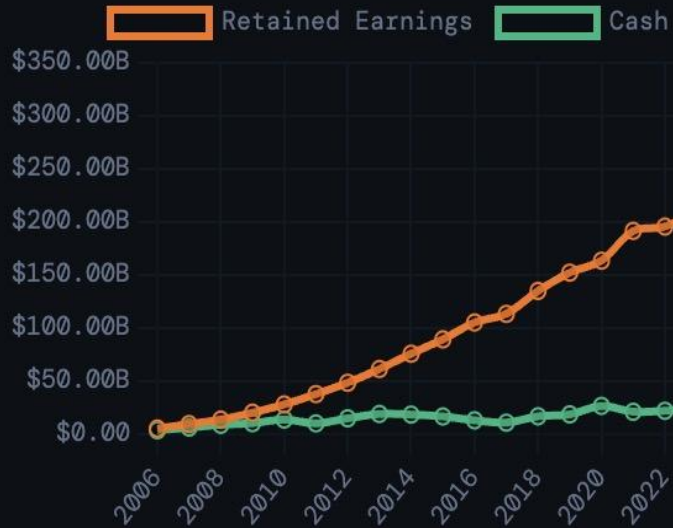
Equity growing strongly.

Low financial risk.

Growth funded mostly internally.

Retained Earnings & Cash

Value accumulation over time



Retained earnings grow steadily and strongly.
Cash relatively stable but increasing slowly.
Strong internal capital base with profit
Company is financially conservative.

Current Ratio

Short-term liquidity



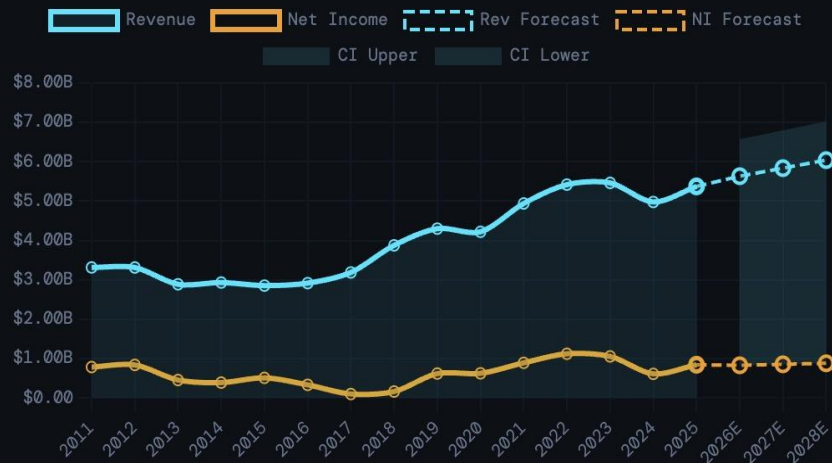
Extremely high in early years (above 8–10).
Gradual decline over time.
Forecast goes very low and would be risky in future
Cash stable but increased slowly

Kesight (KEYS)

Revenue & Net Income

LIN REG + CI

With regression + 90% confidence interval forecast



Revenue :\$3B in 2011 to \$5.5B in 2025
declined slightly -> growth from 2017 to 2023

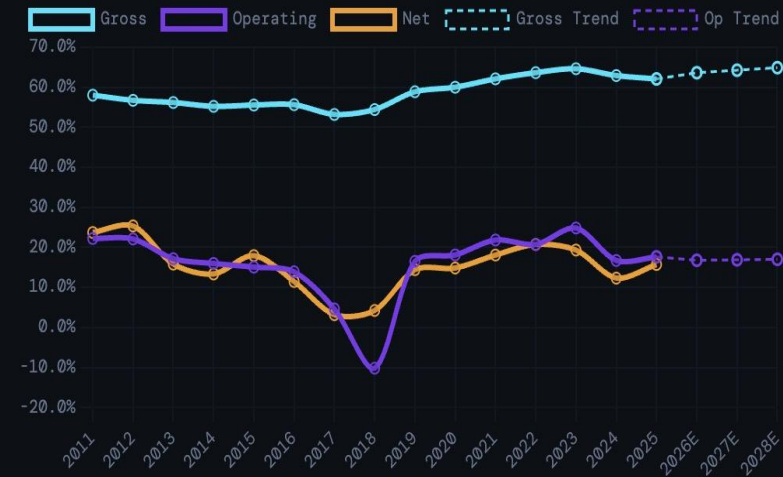
Net income: volatile in early years
Improved after 2019

Profitability is now more stable than earlier years

Profitability Margins

MARGINS

Gross · Operating · Net – with trend lines



Gross Margin: Around 55-65%, very strong and stable

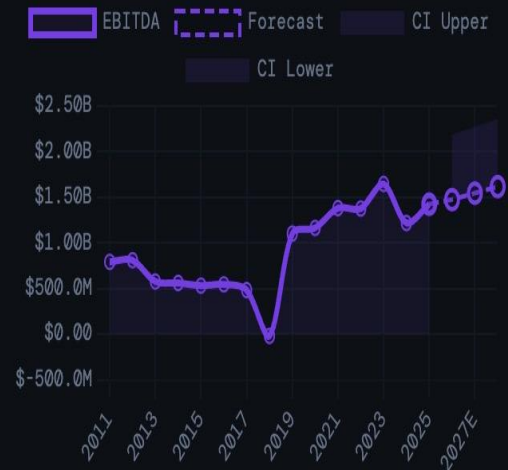
Operating Margin: Dropped sharply around 2017–2018
Recovered strongly after 2019

Net Margin: Similar pattern to operating margin
2017–2018 was a weak period; Profit margins are now stable and healthy

EBITDA

With forecast + CI

EBITDA



Stable but moderate until 2017.
Negative dip around 2018.
Strong increase after 2019
2018 was likely a restructuring or one-time issue.(pandemic)

ROE & ROA

Return on Equity & Assets

RETURNS



ROE:Very high in early years
Drop in 2016–2018 then recovered
ROA:Similar pattern but smoother
Lower than ROE
Slower future growth or higher equity base

R&D Spend

As % of Revenue

R&D



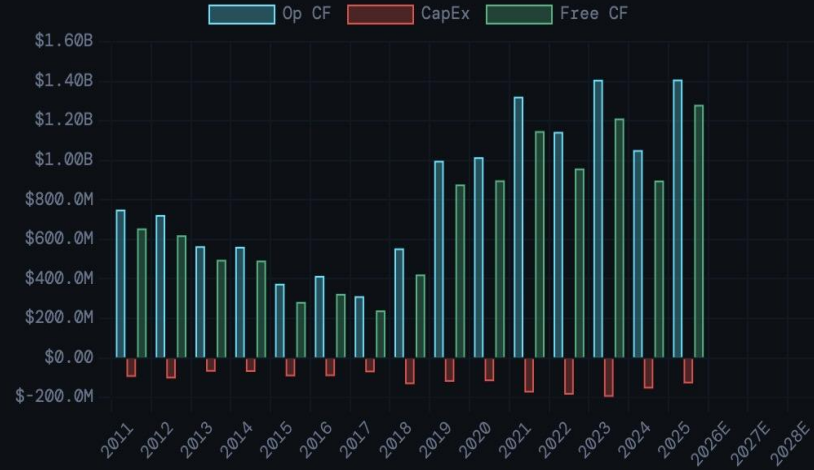
Around 11% in 2011.
Gradual increase over time
Company is increasing investment



Free Cash Flow

Operating CF · CapEx · FCF with forecast + CI

FCF + CI



Cash Flow positive, OPCF increased after 2018

CapEx is small relative to operating cash flow

FCF strong after 2019

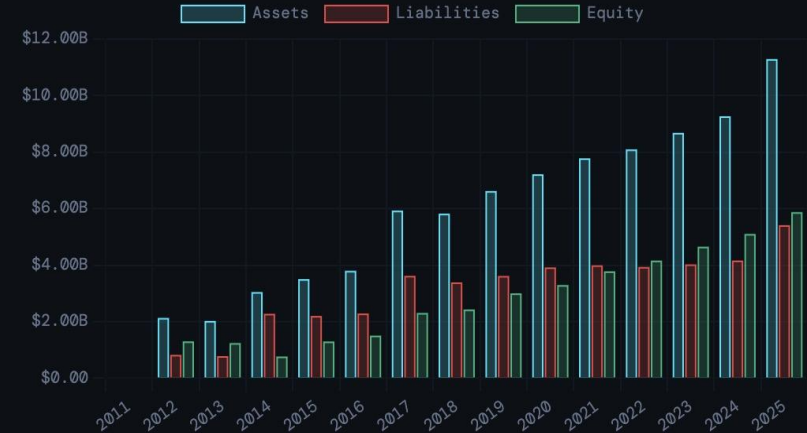
Business generates stable cash with low capital intensity.

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Capital Structure

Assets · Liabilities · Equity

BALANCE SHEET



Assets increasing steadily.

Liabilities increasing but at slower pace.

Equity increasing consistently most of the years

Health growth; No leverage

Stable financial structure.

Current Ratio

Short-term liquidity

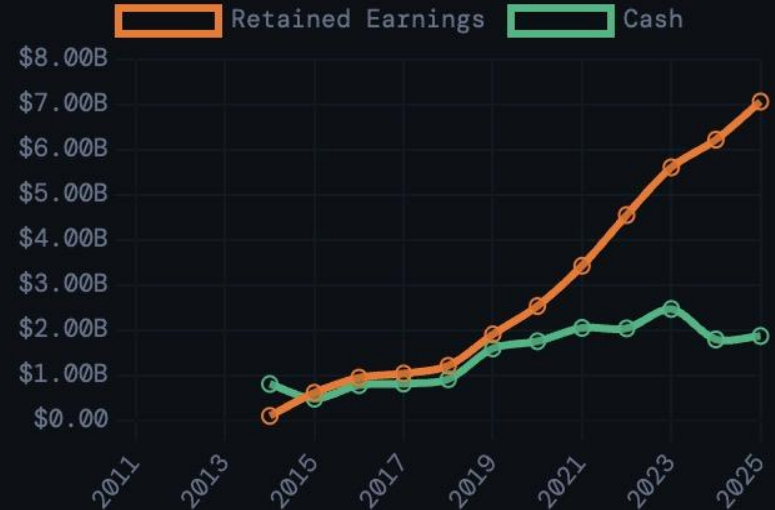


Mostly between 2.3 and 3.2 with strong liquidity.
Forecast remains above 3.0

- Company has strong short-term financial safety.

Retained Earnings & Cash

Value accumulation over time

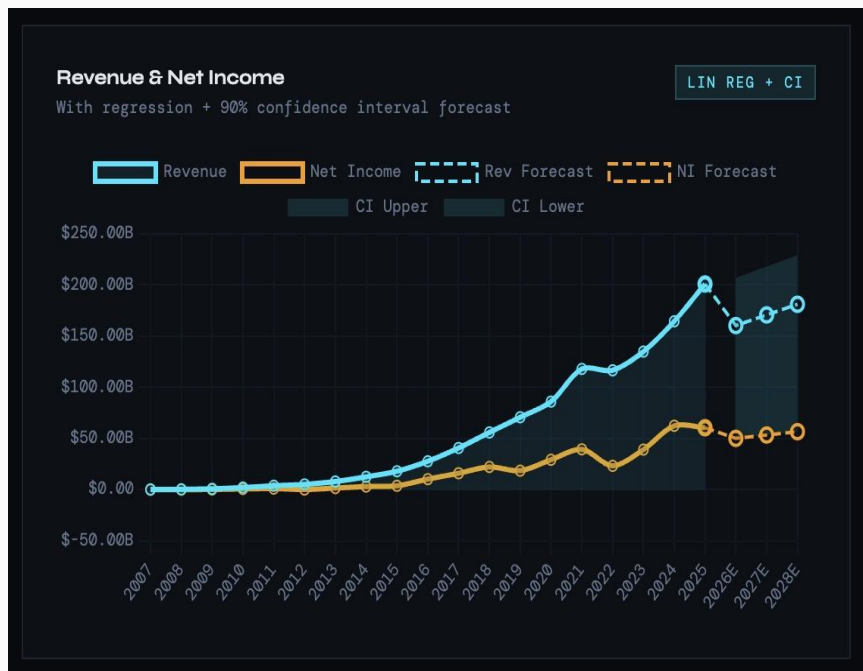


Retained earnings rising steadily.
Cash increasing gradually.

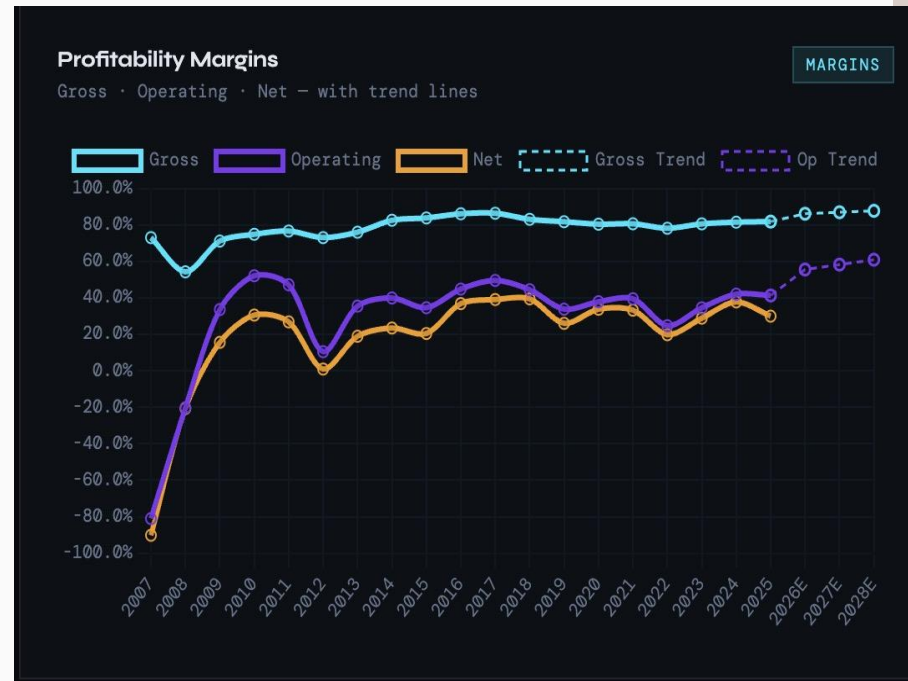
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Profits are being accumulated with Conservative financial management

Meta (META)



Revenue grew from near zero to about \$200B.
 Net income strong growth -> drop -> recovery
 2022 weak
 Business recovered after 2022



Gross Margin: Very high ; stable
 Operating Margin: Very negative with strong improvement
 Forecast rising towards 60
 Net Margin: same trend as Op Margin
 Core business has very high margin, 2022 has high expense

EBITDA

With forecast + CI

EBITDA



Growth with a big jump
Forecast shows moderate growth
Operating performance improved

ROE & ROA

Return on Equity & Assets

RETURNS

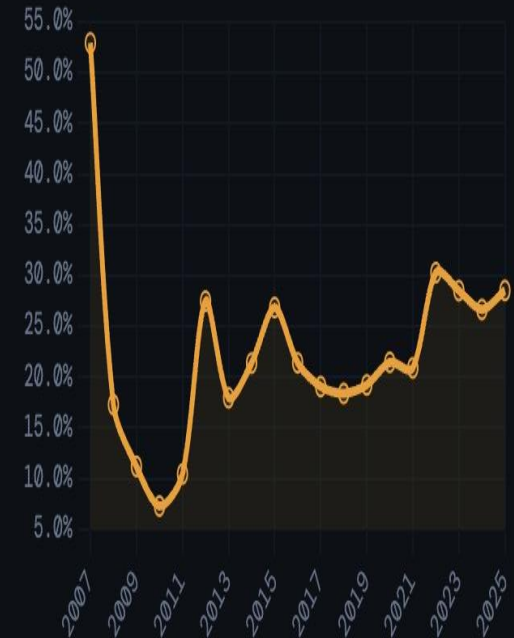


ROE: Drop with a strong rebound stable
forecast
ROA: similar (Good Capital efficiency)

R&D Spend

As % of Revenue

R&D



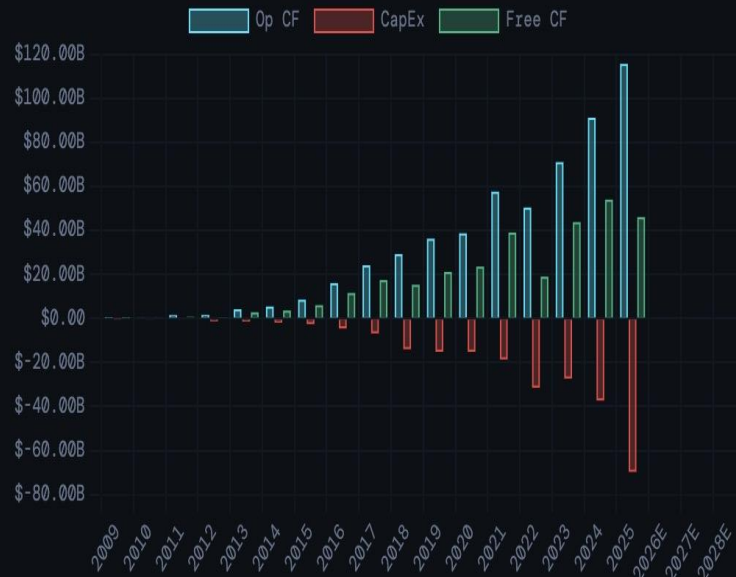
Stabilize recent years
Invest heavily in new projects



Free Cash Flow

FCF + CI

Operating CF · CapEx · FCF with forecast + CI

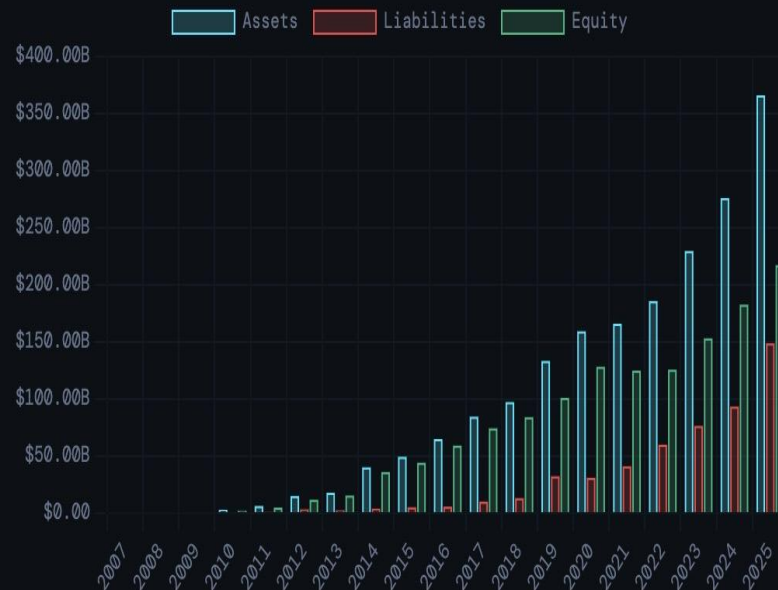


Strong positive FCF after 2013.
Large CapEx increase after 2021.
CapEx spike in 2025
Business generates strong cash.
Recently spending heavily on infrastructure.

Capital Structure

BALANCE SHEET

Assets · Liabilities · Equity



Assets growing rapidly.
Liabilities increasing but much smaller than assets.
Equity increasing steadily.
Strong balance sheet and no extreme leverage

Current Ratio

Short-term liquidity



Extremely high in early years (above 10).
Declining steadily after 2017
Early years: large cash reserves.
Now: more capital deployed.
Liquidity still acceptable but much lower than before.



Retained Earnings & Cash

Value accumulation over time



Retained earnings rising strongly.
Cash increased, especially after 2022.
Slight fluctuation in strong accumulation.
Company keeps a large profit base

Nvidia (NVDA)

Revenue & Net Income

With regression + 90% confidence interval forecast

LIN REG + CI



Revenue: Steady growth from 2016–2020.

Big increase in 2021, Massive surge in 2022–2025

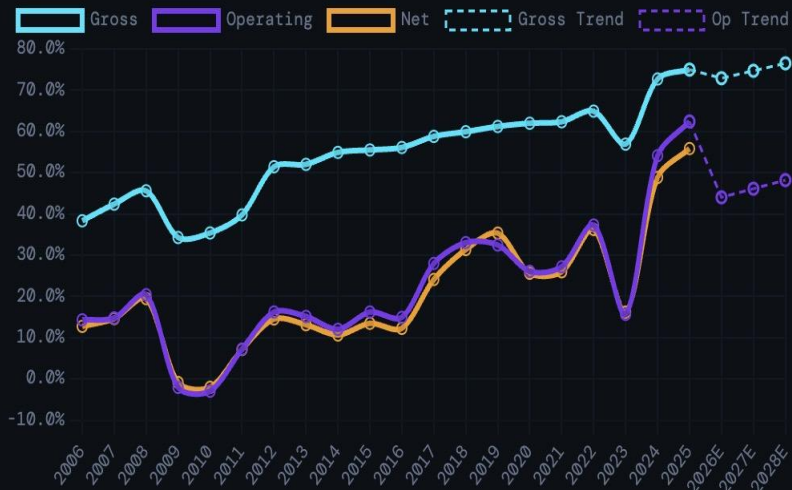
Net Income: Positive growth after 2013. Drop in 2019, 2022.

Explosive growth in 2024–2025 (extraordinary expansion AI)

Profitability Margins

Gross · Operating · Net – with trend lines

MARGINS



GM: Around 40% to 60%, Gradually increased, stable

OPM/NM: weak previously, Very high after 2023

NVIDIA now has extremely high pricing power.

EBITDA

With forecast + CI

EBITDA



Low and flat before 2020
Gradual increase after 2022
Forecast lower than peak but still high
Earnings expansion concentrated recently

ROE & ROA

Return on Equity & Assets

RETURNS

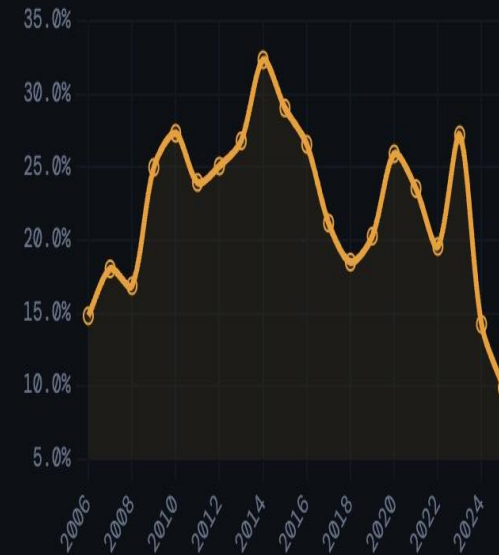


ROE: Mostly 10–30%; Spike above 90% now
ROA: Generally 10–25%. Strong increase
Return on equity is extremely high.
Strong profit generation relative to capital

R&D Spend

As % of Revenue

R&D



Historically between 15%–30%.
Very high during growth years.
Recently declining to around 10%.
Scale effect



Free Cash Flow

FCF + CI

Operating CF · CapEx · FCF with forecast + CI

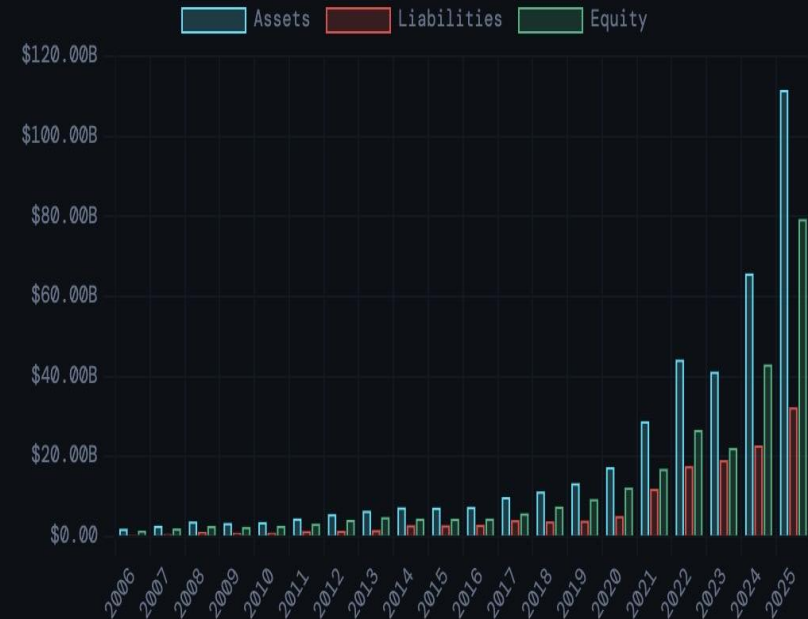


Small before 2016 large jump in 2024–2025
CapEx increasing but too small than cash flow
FCF extremely strong
Cash generation exploded recently. High capital efficiency

Capital Structure

BALANCE SHEET

Assets · Liabilities · Equity



Assets and Equity rising sharply.
Liabilities increasing but much smaller than assets.
Balance sheet expanding.
Growth supported by profits

Current Ratio

Short-term liquidity



Mostly between 3–8; Some volatility
Forecast around 6
Strong liquidity ; No short-term financial stress

Retained Earnings & Cash

Value accumulation over time



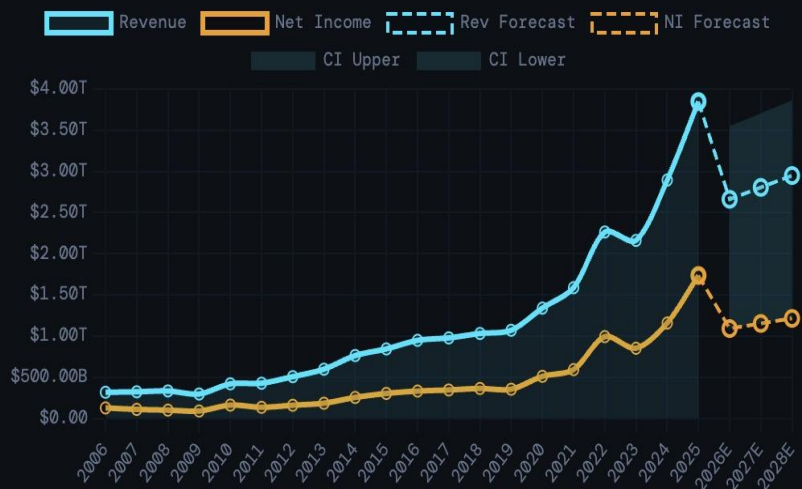
Retained earnings rising sharply after 2023.
Cash fluctuates but increasing overall.
Profits are being retained. Strong capital base.

Taiwan Semiconductor Manufacturing Company (TSM)

Revenue & Net Income

With regression + 90% confidence interval forecast

LIN REG + CI



Revenue: Steady growth from 2006–2016

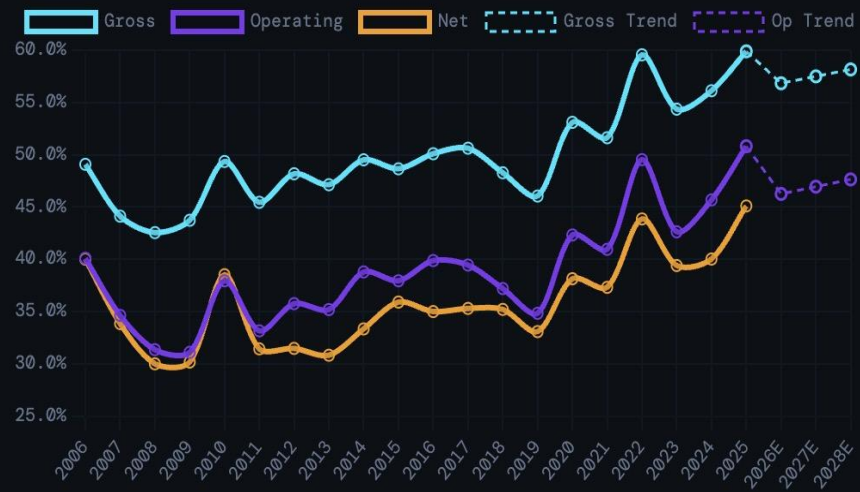
Very strong surge in 2021–2024 (driven by chip demand.)

Net Income: Same trend with lower value

Profitability Margins

Gross · Operating · Net – with trend lines

MARGINS



GM: Mostly between 42%–50%, Fluctuate and increased recently

OPM: Around 30–40% historically Peaked near 50% in 2022.

Net: Around 30–35% long-term. Peaked near mid-40s

Very strong and stable margins. Still structure with high profit

EBITDA

With forecast + CI

EBITDA

EBITDA Forecast CI Upper CI Lower



Smooth upward trend.
Big acceleration recently
Strong operational earnings

ROE & ROA

Return on Equity & Assets

RETURNS

ROE ROA ROE Trend



ROE: Mostly 20–30%.
Peak near mid-30s in recent years
ROA: Mostly 15–20%. Stable.
Strong capital efficiency

R&D Spend

As % of Revenue

R&D



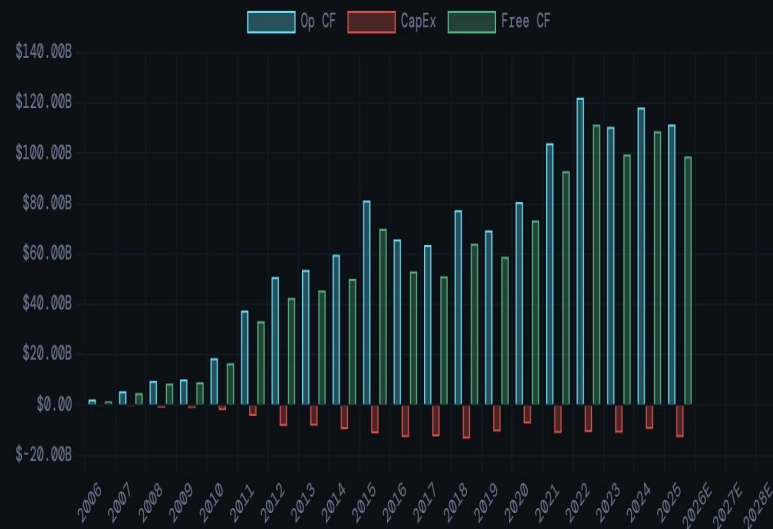
Between 5%–8%.
Slight increase over time.
Recent decline toward 6–7%



Free Cash Flow

Operating CF · CapEx · FCF with forecast + CI

FCF + CI

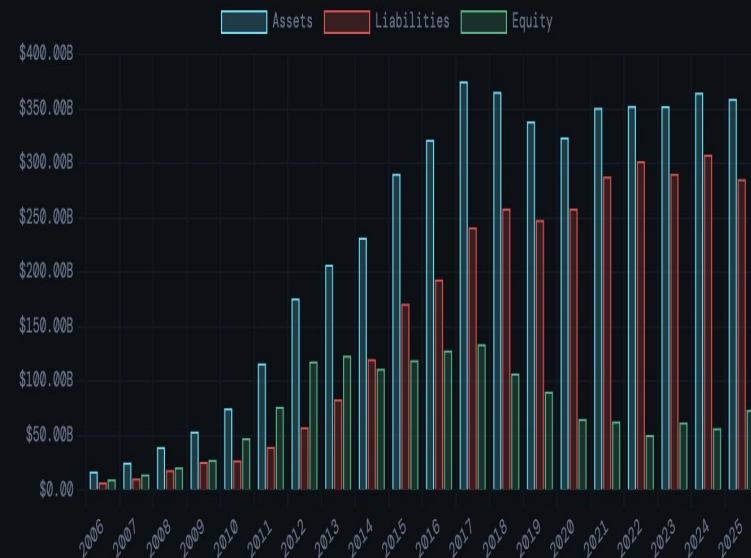


Strong operating cash flow.
Large CapEx every year
Big growth after 2020.
Requires constant large investment.

Capital Structure

Assets · Liabilities · Equity

BALANCE SHEET



Assets growing strongly.
Liabilities increasing.
Equity peaked around 2018 then decreased.
Large capital base Debt higher than earlier years

Current Ratio

Short-term liquidity



Around 2.2 in early years.
Gradually declining. Forecast near 0.5
Liquidity tightening. Less short-term buffer.
More reliance on continuous cash flow.



Retained Earnings & Cash

Value accumulation over time



Retained earnings peaked around 2012–2016.
Then declined sharply to negative
Cash increased moderately
Strong cash but equity structure changed.



05 ✨ Model & Demo ✨

Model

Goal: Identify long-term growth trends in financial data.

Model Used: Linear Regression

$$y_t = \beta_0 + \beta_1 t + \varepsilon_t$$

Where:

- y_t : financial metric (Revenue, Net Income, etc.)
- t : time index
- β_1 : annual growth rate
- ε_t : random noise

```
def linear_regression_with_ci(x, y, forecast_steps=3, alpha=0.10):  
  
    mask = ~(np.isnan(x) | np.isnan(y))  
    xc, yc = x[mask], y[mask]  
    n = len(xc)  
    X = xc.reshape(-1, 1)  
    model = LinearRegression().fit(X, yc)  
    slope = float(model.coef_[0])  
    intercept = float(model.intercept_)  
    y_pred = model.predict(X)  
    r2 = r2_score(yc, y_pred)
```

Validation & Forecasting

- Trend Strength

We are using R^2 to measure how much variation is explained by time trend

```
r2_____ = r2_score(yc, y_pred)
```

- Hypothesis Testing

We compute the t and p value

```
t_stat = slope / se_slope if se_slope > 0 else 0.0  
p_value = 2 * (1 - stats.t.cdf(abs(t_stat), df=n-2))
```

- Forecast with 90% Prediction Interval

```
pi_half = [  
    t_crit * s * math.sqrt(1 + 1/n + (fx - x_mean)**2 / sxx) if sxx > 0 else 0  
    for fx in future_x  
]  
lower = forecast_y - pi_half  
upper = forecast_y + pi_half
```

Model Outputs Overview

```
return {  
  "slope": slope,  
  "intercept": intercept,  
  "r2": r2,  
  "n": n,  
  "t_stat": t_stat,  
  "p_value": p_value,  
  "se_slope": se_slope,  
  "fitted": y_pred.tolist(),  
  "fitted_x": xc.tolist(),  
  "forecast_x": future_x.tolist(),  
  "forecast_y": forecast_y.tolist(),  
  "forecast_lower": lower.tolist(),  
  "forecast_upper": upper.tolist(),  
  "cagr": cagr,  
  "trend": "upward" if slope > 0 else "downward",  
  "significance": "significant" if p_value < 0.05 else "not significant",  
}
```

$$y_t = \beta_0 + \beta_1 t + \varepsilon_t$$

Compound Annual Growth Rate

$$\text{CAGR} = \left(\frac{\text{Ending Value}}{\text{Beginning Value}} \right)^{\frac{1}{T}} - 1$$

Where:

- Ending Value = last value
- Beginning Value = first value
- T = number of years

Demo



06 ★ Conclusion ★



Challenges and Future Work

Challenges:

- Data cleaning
 - Standardization of units/variable names across datasets
 - Consistent format and data types
- Using multiple coding platforms
 - R Studio, HTML, Python
 - Proper storing and use of each platform

Future Work:

- Expand to a macroeconomic lens
 - Add in interest rate, inflation, GDP, etc.
- Zoom in more on specific financial topics
 - Stock market, revenue, etc.
- Look into more companies/predictor variables
 - check if our model still summarizes general trends by adding new influences

Thank You